On classification of SPTs by exact factorizations

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Abstract

In this note, we describe the classification of symmetry-protected topological (SPT) phases based on generalized symmetries, in arbitrary dimensions. We utilize category-theoretic results to obtain a classification which only requires group-theoretical data. Specifically, the classification of these SPT phases boils down to finding exact factorizations of groups and associated cohomology classes. We outline a procedure to realize explicit lattice models of these SPT phases by gauging a Hamiltonian exhibiting spontaneous symmetry breaking (SSB).

Introduction Symmetry-protected topological (SPT) phases of spin chains were first classified throught the lens of cohomology theory in [CGW11]. Subsequently, this classification was extended in [CGLW13] to arbitrary dimensions. SPT phases that break part of the symmetry were first considered in [SPGC11]. The advent of generalized symmetries [GKSW15], specifically non-invertible- and higher-form symmetries, have garnered increasing attention (see e.g. [SS24] [LL24] [FTA25] [GRK23] for lattice realizations). In (1+1)d these generalized symmetries are organized by fusion 1-categories [BT18]. In a similar vein, it is expected that in (2+1)d, generalized symmetries are organized by fusion 2-categories (see e.g [CYL26] [VDCD25] for lattice realizations). When an SPT phase is protected by such a generalized symmetry, we call them *generalized SPTs*. Following the classification of SPT phases protected by conventional on-site group symmetries, we seek to classify also these generalized SPTs in arbitrary dimensions.

[Question to self: definition of fusion 1-category, fiber 1-functor, pointed fusion 1-categories, examples of fusion 1-categories, examples of group-theoretic fusion 1-categories]

Classification of (1+1)d generalized SPTs In this subsection, we attempt to classify $(1+1)$d SPT phases with symmetry encoded in fusion 1-category $\mathcal{C}$. Distinct SPT phases are given by distinct (i.e. non-isomorphic) fiber 1-functors of $\mathcal{C}$ [TW19]. It was shown in [EGNO15] that $\mathcal{C}$ admits a fiber

1-functor (and thus an SPT phase) if and only if $\mathcal{C} \cong \mathbf{Rep}(H)$ for H a finite-dimensional semisimple Hopf algebra. We first introduce the notion of *group-theoretical fusion 1-categories* [ENO17] [NNW09], denoted by $\mathcal{C}(G, \omega; H, \nu)$. Here $H \subseteq G$ is a subgroup of G, ω is a 3-cocycle on G: $\omega \in Z^3(G, U(1))$, and ν is a 2-cochain on H: $\nu \in C^2(H, U(1))$ such that $d\nu = \omega|_H$. Physically, the group-theoretical fusion 1-category $\mathfrak{C}(G, \omega; H, \nu)$ describes the symmetry of a model with symmetry G defined on a chain, and applying a (twisted) gauging of H [LS23]. Importantly, the fiber 1-functors (and thus $(1+1)$d SPT phases) are described by group- and cohomology-theoretical data [Ost01] [Nat17], such that their classifcation does not necessitate knowledge of abstract category theory. We refer to [Nat17] for the classification of $(1+1)$d SPT phases with group-theoretic fusion 1-category $\mathfrak{C}(G, \omega; H, \nu)$. (Un)fortunatly, not all fusion 1-categories admitting a fiber 1-functor are group-theoretical fusion 1-categories [Nik07], and classifying finite-dimensional semisimple Hopf algebra remains an open problem [DY25]. Thus, we cannot provide a comprehensive classification of all $(1+1)$d SPT phases. This is to be contrasted with the categorified version of group-theoretical fusion 1-categories, where it is found that all fusion 2-categories admitting a fiber 2-functor are group-theoretical fusion 2-categories [DY25], a fact elaborated upon in next subsection.
We shall henceforth focus on group-theoretic fusion 1-categories $\mathfrak{C}(G, \omega; H, \nu)$. The discussion on the classification of $(1+1)$d SPT phases under this type of symmetry parallels that of the classification of $(2+1)$d SPT phases, such that we allude to the next subsection for the classifcation of phases under this type of symmetry.

Classification of (2+1)d generalized SPTs In this subsection we attempt to classify $(2+1)$d SPT phases with symmetry encoded in fusion 2 category $\mathcal{C}$. Looking ahead, we will find that we can fully classify all (2+1)d SPT phases using only group- and cohomology theory. We first introduce the notion of *group-theoretic fusion 2-categories*, denoted $\mathfrak{C}(G, \omega; H, \nu)$, which is the strong categorification of *group-theoretical fusion 1-categories* explictly introduced in [ENO17]. Here, G is a finite group symmetry with (possibly anomalous) 4-cocycle $\omega \in Z^4(G, U(1))$, and $H \subseteq G$ is a non-anomalous subgroup with discrete torsion given by the 3-cochain $\nu \in C^3(H, U(1))$ such that $d\nu = \omega|_H$ [DY25] [*difference between Decoppét and Inamura]. These fusion 2-categories arise upon applying the (twisted) gauging of the subgroup $H \subseteq G$ of 0-form finite group symmetries in (2+1)d [IO26b]. In the case that both ω and ν are trivial, we denote the 2-category by $\mathfrak{C}(G; H)$. Below we give some examples of *group-theoretic fusion 2-categories*.

Example 1. The simplest example of such a group-theoretic fusion 2-category is obtained by considering $\mathfrak{C}(G, \omega; \{e\}, triv) \cong \mathbf{2Vec}_G^\omega$. Specifically, when $\omega \sim triv$, we have $\mathfrak{C}(G, triv; \{e\}, triv) \cong \mathbf{2Vec_G}$.

Example 2. When gauging the full group G, we have the following 2-category: $\mathfrak{C}(G, triv; G, triv) \cong \mathbf{2Rep}(G)$

Example 3. Given a finite group G_0, consider the group-theoretic category obtained by taking $G_0 \times G_0$ and gauging the $G_0^L := \{(g,e)|g \in G_0\}$: $\mathfrak{C}(G_0 \times G_0; G_0^L) \cong \mathbf{2Rep}(G_0) \boxtimes \mathbf{2Vec}_{G_0}$.

Example 4. As a concrete example, consider the dihedral group of order 8, D_4, which describe the symmetries of the square. Choosing $H = \mathbb{Z}_2$, we get: $\mathfrak{C}(D_4; \mathbb{Z}_2) \cong \mathbf{2Rep}((\mathbb{Z}_2^{(1)} \times \mathbb{Z}_2^{(1)}) \rtimes \mathbb{Z}_2^{(0)})$. This 2-category is also an example of a Tambara-Yamagami 2-category [DY25].

Next, we introduce the notion of *fiber 2-functors of a fusion 2-category*:

Definition 1. Let $\mathfrak{C}$ be a fusion 2-category. A fiber 2-functor for $\mathfrak{C}$ is a monoidal 2-functor $\mathfrak{C} \to \mathbf{2Vec}$.

Physically, we associate a (2+1)d bosonic $\mathcal{C}$-SPT phase to each distinct (isomorphism class of a) fiber 2-functor of the fusion 2-category $\mathcal{C}$ [IO26a] [OI26] [IO26b]. Importantly, it was shown in [DY25] that the only fusion 2-categories that admit such fiber 2-functors are neccessarily also group-theoretical fusion 2-categories [DY25]:

Theorem 1. *A fusion 2-category admits a fiber 2-functor iff it is equivalent to some group-theoretical fusion 2-category $\mathfrak{C}(G, \omega; H, \nu)$ such that for $H \subseteq G$ $\exists K : G = HK$ with $H \cap K = \{e\}$ and $\omega|_K$ trivializable.*

Thus, given some SPT phase, we know that the symmetry protecting this phase is described by a group-theoretical fusion 2-category. A classification of all bosonic (2+1)d SPT phases can thus be achieved by considering only group-theoretical fusion 2-categories and their associated fiber 2-functors. Regarding the labelling of distinct SPT phases, we focus on the conditions on K given in theorem 1. To this end, we first introduce *exact factorizations of a group*:

Definition 2. Let G be a finite group. An exact factorization of G is given by two subgroups $H, K \subseteq G$ such that $G = HK$, and with trivial intersection: $H \cap K = \{e\}$, where e denotes the trivial element of G. For a fixed subgroup $H \subseteq G$, we call the subgroup $K \subseteq G$ the complement of H in G.

Hence, the labelling of distinct SPT phases protected by the symmetry encoded in $\mathfrak{C}(G, \omega; H, \nu)$ is specified by the pair (K, λ) where K is the complement of H in G, and $\lambda \in Z^3(K, U(1)) : d\lambda = \omega|_K$ [DY25][IO26b]. The equivalence between phases labelled by (K, λ) and (K', λ') is given by the existence of a $g \in G$ such that the adjoint action of g relates K to K', and a parallel condition relating λ and λ' [DY25].

Example 5. Consider the group-theoretical fusion 2-category introduced in example 3 given by $\mathfrak{C}(G_0 \times G_0; G_0^L) \cong \mathbf{2Rep}(G_0) \boxtimes \mathbf{2Vec}_{G_0}$. It was shown in [IO26b] that distinct phases (given $\lambda = triv$) are labelled by $K(f) := \{(f(g), g)|g \in G_0\}$. Equivalently, the phases can thus be labelled by the endomorphisms $f \in \mathrm{End}(G_0)$, up to equivalence given by the adjoint action of some $g_l \in G_0^L$: $f' = \mathrm{Ad}_{g_l} \circ f$.

Example 6. The group-theoretical fusion 2-category $\mathfrak{C}(D_4;\mathbb{Z}_2) \cong \mathbf{2Rep}((\mathbb{Z}_2^{(1)} \times \mathbb{Z}_2^{(1)}) \rtimes \mathbb{Z}_2^{(0)})$ introduced in example 4 admits two distinct SPT phases labelled by the complements $K_1 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong V_4$ and $K_2 \cong \mathbb{Z}_4$, given that $\lambda = triv$ for both phases.

At this point we answer the question on how to construct lattice models which describe the $\mathfrak{C}(G,H)$-SPT phase labelled by (K,λ). This is done by starting from a Hamiltonian with spontaneous symmetry breaking (SSB) pattern $G \to K$ wherein $\lambda \in C^3(K,U(1))$ describes the SPT class of K, and subsequently gauging H.

Classification of (n+1)d generalized SPTs In this subsection we attempt to classify $(n+1)$d SPT phases in arbitrary spatial dimensions n, protected by *fusion n-categories* [KWZ15]. Following the previous two subsections on the one- and two-dimensional cases, we will restrict to *group-theoretical fusion n-categories*, denoted $\mathfrak{C}(G,\omega;H,\nu)$. Here $H \subseteq G$ denotes a subgroup of G, ω is a $(n+2)$-cocycle on G: $\omega \in Z^{n+2}(G,U(1))$, and ν is a $(n+1)$-cochain on H: $\nu \in C^{n+1}(H,U(1))$ satisfying $d\nu = \omega|_H$. These so-called group-theoretic fusion n-categories are obtained by starting with some model in n spatial dimensions with symmetry G, and subsequently applying a (twisted) gauging of H. We again expect non-isomorphic *fiber n-functors* of $\mathfrak{C}(G,\omega;H,\nu)$ to correspond to distinct $(n+1)$d SPT phases. The distinct SPT phases are then labelled by subgroups K satisfying the exact factorization condition, together with a $(n+1)$-cocycle λ on K, i.e. $(n+1)$d SPT phases protected by a group-theoretic fusion n-category are labeled by the pairs (K_i,λ_{i_j}).

Summary Although the results presented above rely on the abstract mathematical framework given by (higher) category theory, the quest of constructing explicit lattice models of generalized SPTs based on a group-theoretic fusion n-category $\mathfrak{C}(G;H)$ boils down to a group theory problem. Specifically, given a group G and the subgroup $H \subseteq G$, one finds all possible $\mathfrak{C}(G;H)$-SPT phases by enumerating all possible complements $\{K_i\}$ of H in G, and the possible $(n+1)$-cocycles λ_{i_j} on each of the K_i: $\lambda_{i_j} \in Z^{n+1}(K_i,U(1))$. The corresponding lattice realization of an SPT labelled by (K_i,λ_{i_j}) is obtained by first writing down a Hamiltonanian which breaks a symmetry from G to K_i and $\lambda_{i_j} \in H^{n+1}(K_i,U(1))$ specifies the SPT class of K_i before gauging, and subsequently gauging the subgroup $H \subseteq G$ [IO26b]. The gauging of a (sub)symmetry can be found in [HVAS+15] [GRK23] [CGRV+25] [BT18] [VDCD25]. We thus have an operational procedure to write down explicit lattice models of generalized-SPTs [IO26b] [IHTSN26].
Equipped with the results regarding the classification of $(1+1)$d- and $(2+1)$d-SPTs, we believe the following questions are interesting:

- Why can we achieve a comprehensive classification of all SPTs in $(2+1)$d in terms of group- and cohomology theory, but the same cannot be said for $(1+1)$d SPTs?

- For what spatial dimensions n, are all SPTs based on a group-theoretic fusion n-category?
- How does one write down a model of an SPT **not** described by a group-theoretic fusion category?

Answering the first question, might give a helpful clue on the second question. For the third question, we would like to specifically write down lattice modes and to use the tensor network framework [CPGSV21] to investigate such phases.

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